

The type-(B) compact self-commutator problem

A candidate full solution to Question 4.10 of arXiv:1303.4844

Automated mathematics run `fa_banach_001`, lane 0

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Status. Candidate full solution, likely valid; human review requested. The proof is self-contained apart from the compact spectral theorem and the standard real orthogonal normal form for real skew-symmetric matrices.

Abstract

Let J be a conjugation on a separable infinite-dimensional complex Hilbert space H , and let

$$\mathfrak{o}_J = \{X \in \mathcal{K}(H) : X = -JX^*J\}$$

be the type-(B) compact orthogonal Lie algebra. We prove that every self-adjoint $T \in \mathfrak{o}_J$ is a self-commutator $T = [Y^*, Y]$ of an element $Y \in \mathfrak{o}_J$. In fact Y can be chosen with $\|Y\| \leq \sqrt{2}\|T\|^{1/2}$. Consequently the possible eigenvalue multisets are exactly the real null multisets in which every nonzero λ and $-\lambda$ have equal multiplicity. The compactness issue is resolved by grouping two spectral pairs at a time into four-dimensional orthogonal blocks; an explicit square-root estimate makes their direct sum compact.

1 Source problem

The source is Beltiță, Patnaik, and Weiss, *B(H)-Commutators: A Historical Survey II and recent advances on commutators of compact operators*, arXiv:1303.4844 [1]. Definition 4.1 introduces the type-(B) algebra $\mathfrak{o}_\infty(H)$. Question 4.10 on PDF page 13 asks:

“Is it possible to establish versions of Theorem 4.6 for the complex classical Lie algebras of types (B) and (C) introduced in Definition 4.1?”

Here Theorem 4.6 is the Fan–Fong spectral characterization of compact self-commutators in type (A). Immediately before the question, on PDF page 12, the authors say that the type-(B) problem is completely open and is not addressed in the paper. The source passage is reproduced below.

Some motivation for introducing the terminology in Definition 4.3 is provided by Question 4.10 below, which basically asks for identifying the sequences of real numbers which can occur as eigenvalue sequences for self-commutators in the complex classical Lie algebras of type (A), (B), (C). It would be natural to call these sequences of types (A), (B), (C), respectively, but such a definition has the drawback that it is very implicit. For the sequence of type (A) we prefer a different approach in Definition 4.3, and we further clarify our reasons as follows.

The problem of identifying the eigenvalue sequences for Lie algebras of type (A) was solved in [10] (see Theorem 4.6 below). For algebras of type (C), this problem will be solved in Theorem 4.12 below. The problem for classical Lie algebras of type (B) is completely open. We do not address it in this paper and we will not propose any explicit definition for the sequences of type (B).

The source subsequently gives a complete answer for type (C), but explicitly not for type (B). The result below supplies the missing type-(B) version.

2 Notation and elementary consequences

Throughout, H is a separable infinite-dimensional complex Hilbert space, J is a conjugation, and

$$\mathfrak{o}_J := \{X \in \mathcal{K}(H) : X = -JX^*J\}.$$

For matrix units, write $L_{jk} = E_{jk} - E_{kj}$. Thus L_{jk} is real skew-symmetric and $L_{jk}^* = -L_{jk}$.

If $T = T^* \in \mathfrak{o}_J$, then

$$T = -JTJ, \quad TJ = -JT. \quad (2.1)$$

Hence J maps the λ -eigenspace antiunitarily onto the $(-\lambda)$ -eigenspace. In particular, nonzero eigenvalues occur in opposite pairs with equal multiplicity.

On a finite-dimensional J -invariant subspace, choose a J -real orthonormal basis. In that basis J is coordinatewise conjugation and the relation $X = -JX^*J$ becomes

$$X^T = -X. \quad (2.2)$$

Thus the relevant finite blocks are ordinary complex skew-symmetric matrices.

3 Proof intuition

A single spectral pair $\{\lambda, -\lambda\}$ lives in a two-dimensional J -invariant space, but $\mathfrak{so}(2, \mathbb{C})$ is abelian, so it cannot produce a nonzero self-commutator. The obstruction disappears as soon as two pairs are combined: an explicit $\mathfrak{so}(4, \mathbb{C})$ formula produces both pairs at once. If a finite-rank operator leaves one pair over, its infinite-dimensional kernel supplies a J -fixed zero vector and an $\mathfrak{so}(3, \mathbb{C})$ formula handles the resulting block.

The important quantitative feature is not merely finite-dimensional solvability. The formulas give $\|Y_k\| \leq \sqrt{2}\|T_k\|^{1/2}$ on every block. Since the eigenvalues of compact T tend to zero, the norms of the block solutions tend to zero. Their orthogonal direct sum is therefore compact and remains in the same type-(B) algebra.

4 Finite block formulas

Lemma 1 (three- and four-dimensional blocks). *Let $m \in \{3, 4\}$ and let $S \in M_m(\mathbb{C})$ satisfy $S = S^*$ and $S^T = -S$. Then there is $Z \in M_m(\mathbb{C})$ such that*

$$Z^T = -Z, \quad [Z^*, Z] = S, \quad \|Z\| \leq \sqrt{2}\|S\|^{1/2}. \quad (4.1)$$

Proof. Since $S = S^*$ and $S^T = -S$, we have $\bar{S} = -S$, so $S = iR$ for a real skew-symmetric matrix R . The real orthogonal normal form for R shows that, after conjugating by a real orthogonal matrix, the target is

$$i\lambda L_{12} \quad (m=3), \quad i\lambda L_{12} + i\mu L_{34} \quad (m=4), \quad (4.2)$$

where $\lambda \geq \mu \geq 0$. Real orthogonal conjugacy preserves transpose, adjoint, commutators, and operator norm, so it suffices to solve these forms.

For $m = 3$, use $[L_{13}, L_{23}] = -L_{12}$ and set

$$Z = \sqrt{\frac{\lambda}{2}}(L_{13} + iL_{23}). \quad (4.3)$$

Then $Z^T = -Z$ and

$$[Z^*, Z] = -2i\frac{\lambda}{2}[L_{13}, L_{23}] = i\lambda L_{12}.$$

Because $\|L_{jk}\| = 1$, the triangle inequality gives $\|Z\| \leq \sqrt{2\lambda}$.

For $m = 4$, put

$$\begin{aligned} A_1 &= (L_{12} + L_{34})/2, & A_2 &= (L_{13} - L_{24})/2, & A_3 &= (L_{14} + L_{23})/2, \\ B_1 &= (L_{12} - L_{34})/2, & B_2 &= (L_{13} + L_{24})/2, & B_3 &= (L_{14} - L_{23})/2. \end{aligned} \quad (4.4)$$

Direct multiplication gives

$$[A_2, A_3] = -A_1, \quad [B_2, B_3] = B_1, \quad [A_j, B_k] = 0. \quad (4.5)$$

Define

$$Z = \sqrt{\frac{\lambda + \mu}{2}}(A_2 + iA_3) + \sqrt{\frac{\lambda - \mu}{2}}(B_2 - iB_3). \quad (4.6)$$

Every summand is skew-symmetric. Equations (4.5), including the vanishing of all cross terms, yield

$$\begin{aligned} [Z^*, Z] &= i(\lambda + \mu)A_1 + i(\lambda - \mu)B_1 \\ &= i\lambda L_{12} + i\mu L_{34}. \end{aligned}$$

The operators A_j and B_j have norm $1/2$, so $\|A_2 + iA_3\| \leq 1$ and $\|B_2 - iB_3\| \leq 1$. Therefore

$$\begin{aligned} \|Z\| &\leq \sqrt{\frac{\lambda + \mu}{2}} + \sqrt{\frac{\lambda - \mu}{2}} \\ &\leq \sqrt{2\lambda} = \sqrt{2}\|S\|^{1/2}. \end{aligned}$$

This proves the lemma. □

5 Main theorem

Theorem 2 (type-(B) self-commutator theorem). *For every self-adjoint $T \in \mathfrak{o}_J$ there exists $Y \in \mathfrak{o}_J$ such that*

$$[Y^*, Y] = T \quad \text{and} \quad \|Y\| \leq \sqrt{2}\|T\|^{1/2}. \quad (5.1)$$

Proof. The assertion is immediate for $T = 0$, so assume $T \neq 0$. By the compact self-adjoint spectral theorem and (2.1), choose an orthonormal list (u_n) of eigenvectors for the positive eigenvalues, repeated with multiplicity, so that

$$Tu_n = \lambda_n u_n, \quad T(Ju_n) = -\lambda_n Ju_n, \quad \lambda_n > 0. \quad (5.2)$$

The two vectors in each pair are orthogonal because they belong to distinct eigenspaces. If the list is infinite, arrange it so that $\lambda_n \downarrow 0$.

For each n define

$$f_n = \frac{u_n + Ju_n}{\sqrt{2}}, \quad g_n = \frac{i(u_n - Ju_n)}{\sqrt{2}}. \quad (5.3)$$

Then f_n, g_n are orthonormal and fixed by J . On their span, in the ordered basis (f_n, g_n) , the matrix of T is $i\lambda_n L_{12}$.

Suppose first that there are infinitely many positive eigenvalue pairs. Group the pairs with indices $2k-1, 2k$ into mutually orthogonal four-dimensional J -invariant spaces H_k . Relabeling within a block if necessary, Lemma 1 gives a skew-symmetric block Y_k with

$$[Y_k^*, Y_k] = T|_{H_k}, \quad \|Y_k\| \leq \sqrt{2} \max(\lambda_{2k-1}, \lambda_{2k})^{1/2}. \quad (5.4)$$

The right-hand side tends to zero.

If there are finitely many positive eigenvalue pairs, group them in the same way. When their number is odd, one pair remains. In that case T has finite rank, hence $\ker T$ is infinite-dimensional and J -invariant. The restriction of J to $\ker T$ has a fixed unit vector h : for a nonzero $x \in \ker T$, either $x + Jx$ is nonzero and can be normalized, or ix is J -fixed after normalizing. Adjoin h to the remaining pair and apply the three-dimensional part of Lemma 1. This completes the finite block decomposition in all cases.

Let H_0 be the orthogonal complement of the constructed blocks. It is a J -invariant subspace of $\ker T$. Define

$$Y = 0|_{H_0} \oplus \bigoplus_k Y_k. \quad (5.5)$$

Each block is skew-symmetric in a J -real basis, so (2.2) gives $Y = -JY^*J$. The block norms tend to zero in the only infinite case; therefore the direct sum in (5.5) is compact. Thus $Y \in \mathfrak{o}_J$. Finally, blockwise computation gives

$$[Y^*, Y] = 0|_{H_0} \oplus \bigoplus_k [Y_k^*, Y_k] = T.$$

Taking the supremum in the block estimate proves (5.1). $\square$

Corollary 3 (spectral characterization). *A real null multiset is the eigenvalue multiset of a self-commutator $[Y^*, Y]$ with $Y \in \mathfrak{o}_J$ if and only if every nonzero λ and $-\lambda$ occur with equal multiplicity. Equivalently, after relabeling, the nonzero terms have the form*

$$\lambda_1, -\lambda_1, \lambda_2, -\lambda_2, \dots, \quad \lambda_n > 0, \text{quad} \lambda_n \longrightarrow 0, \quad (5.6)$$

with any allowed zero multiplicity.

Proof. Necessity follows from (2.1), because every $[Y^*, Y]$ is self-adjoint and belongs to the Lie algebra $\mathfrak{o}_J$. Conversely, realize (5.6) by the compact block-diagonal operator whose two-dimensional J -real blocks are $i\lambda_n L_{12}$, and apply Theorem 2. $\square$

6 Verification and novelty limits

The identities in Lemma 1 were checked both symbolically and in 1,000 randomized orthogonally conjugated instances; see `code/verify_blocks.py` and `verification.md`. These checks are sanity tests, not a substitute for the displayed algebraic proof.

A bounded novelty check was performed on 19 July 2026. The run’s registry, solution, attempt, and proof-gap indexes were searched for arXiv:1303.4844, the exact title, “type B”, “orthogonal Lie algebra”, and “self-commutator”. Primary-source arXiv/web searches used the exact source question, the phrase “type (B) self-commutator”, the equation $[X^*, X] = A$ in the compact orthogonal algebra, the source authors, and citing/structural work on the orthogonal operator Lie algebra. No later

paper explicitly answering this type-(B) question was located. This is a bounded search, not a claim of exhaustive historical priority.

The theorem addresses only the type-(B) part of Question 4.10. It does not address the source paper's separate questions about positive single commutators of compact operators, distinct eigenvalues, or block-size control.

Human review recommendation. Check (i) the signs and cross-term cancellation in (4.3)–(4.6), (ii) the construction of a J -real spectral basis, (iii) the odd finite-rank kernel-vector case, and (iv) the compact direct-sum argument. No unproved mathematical dependency remains in the candidate proof.

References

- [1] D. Beltiță, S. Patnaik, and G. Weiss, *B(H)-Commutators: A Historical Survey II and recent advances on commutators of compact operators*, arXiv:1303.4844 (2013), especially Definition 4.1, Question 4.10, and the discussion on pp. 12–13.